Arista Access Network QoS Congestion Event Resolution - MALL-ACC-80

Node: MALL-ACC-80 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_4147

Description

This Standard Operating Procedure outlines the diagnostic and resolution steps for a QoS congestion event detected on the Arista Access network equipment located at the Malleswaram Node, Node ID: MALL-ACC-80. The failure domain of interest is related to microbursts, interface tail drops, and QoS queue starvation, which can lead to reduced network performance and Quality of Service (QoS) degradation. On Arista Access network equipment, QoS is primarily managed through the use of egress queues, which are implemented using the 8-tuple match and the egress queue policy. The egress queue policy is used to specify the egress queue to use for a particular traffic flow. The egress queue policy is also used to specify the buffer size and the drop policy for the egress queue. If the egress queue buffer is full and packets continue to arrive, tail drops will occur, leading to packet loss and QoS degradation. Additionally, microbursts can occur when a high volume of packets are transmitted, causing the interface to become congested and leading to QoS degradation. The QoS congestion event can be caused by various factors, including, but not limited to, misconfigured egress queues, insufficient buffer sizes, and high traffic volumes. It is essential to thoroughly investigate the root cause of the QoS congestion event to ensure that the resolution steps are effective and do not introduce any unintended consequences. This SOP provides a structured approach to diagnosing and resolving QoS congestion events on Arista Access network equipment.

Pre-requisite Checks

- check: Verify the redundant link status at the Malleswaram Node
- check: Dispatch remote hands to the Malleswaram Node for assistance
- check: Ensure that the Arista Access network equipment is properly powered and configured

Potential Root Causes

Cause	Probability
Misconfigured egress queues	High
Insufficient buffer sizes	Medium
High traffic volumes	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface MALL-ACC-80 detail, Expected Output: Interface MALL-ACC-80, Hardware: 1, Description: This command is used to display detailed information about the interface, including the egress queue configuration and the buffer size.
```

```
Command: show qos policy MALL-ACC-80 detail, Expected Output: Policy MALL-ACC-80, Queue: 1, Description: This command is used to display detailed information about the QoS policy, including the egress queue policy and the buffer size.
```

Resolution Steps

Step 1: Step: Verify the egress queue configuration, Command: show interface MALL-ACC-80 detail, Description: Verify that the egress queue configuration is correct and that the buffer size is sufficient to handle the traffic volume.

Step 2: Step: Adjust the egress queue buffer size, Command: interface MALL-ACC-80, Description: Adjust the egress queue buffer size to a value that is sufficient to handle the traffic volume.

Step 3: Step: Verify the QoS policy configuration, Command: show qos policy MALL-ACC-80 detail, Description: Verify that the QoS policy configuration is correct and that the egress queue policy is properly configured.

Step 4: Step: Adjust the QoS policy configuration, Command: qos policy MALL-ACC-80, Description: Adjust the QoS policy configuration to ensure that the egress queue policy is properly configured.

Post-Check Verification

```
Command: show interface MALL-ACC-80 detail, Expected Output: Interface MALL-ACC-80, Hardware: 1, Description: Verify that the egress queue configuration and buffer size are correct.

Command: show qos policy MALL-ACC-80 detail, Expected Output: Policy MALL-ACC-80, Queue: 1, Description: Verify that the QoS policy configuration is correct.
```

Roll-back Procedures

- Step: Restore the original egress queue configuration, Command: interface MALL-ACC-80, Description: Restore the original egress queue configuration and buffer size.

- Step: Restore the original QoS policy configuration, Command: qos policy MALL-ACC-80, Description: Restore the original QoS policy configuration.